

Additive (Incremental) Bayesian Updating for a BNN — Oil & Gas Field Expansion

Domain: Oil & Gas — production forecasting during field development

Reference: [Turing]]

The question this notebook tests: when new wells are drilled after a model is already in production use, can we *add* their data to the existing posterior — instead of re-running MCMC over the entire well history every time a handful of new wells shows up?

We compare three models, all sharing the same Bayesian-neural-network architecture as '01_bnn_permeability_regression.jl':

1. **Stale** — trained once on Phase A wells only, never updated.
2. **Additive update** — Phase A's posterior is turned into an informative prior (moment-matched Gaussian) and NUTS runs *again only on the new Phase B data*, i.e. sequential Bayesian updating ($p(\theta | A, B) \approx p(\theta | B) \cdot p(\theta | A)$ used as prior).
3. **Full retrain** — NUTS run from scratch on Phase A u Phase B pooled together — the expensive gold standard the additive update is trying to approximate.

```
1 # Additive (Incremental) Bayesian Updating for a BNN — Oil & Gas Field Expansion
2
3 ##Domain## Oil & Gas — production forecasting during field development
4
5 ##Reference## [Turing.jl](https://turinglang.org/)
6
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9 model is already in production use, can we add their data to the
10 existing posterior — instead of re-running MCMC over the entire well
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Additive (Incremental) Bayesian Updating for a BNN — Oil & Gas Field Expansion

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7. Posterior predictive on a Phase B well: stale vs. additive vs. retrain
8. Discussion & caveats
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1. Synthetic production data: two field-development phases

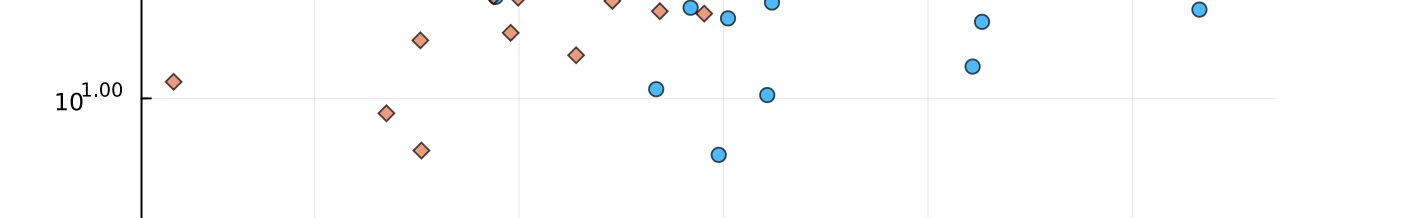
We predict **Estimated Ultimate Recovery**, EUR (thousand barrels), from three early-life well attributes: initial rate q_i (bbl/d), initial Arps decline rate D_i (1/yr) and reservoir permeability k (mD).

- **Phase A ("core area")** — the first 40 wells drilled, used to build the initial model.
- **Phase B ("step-out area")** — 25 wells drilled later in an adjacent fault block with measurably different reservoir quality. This is a deliberate *distribution shift*: it is exactly the situation where a stale model needs updating, and where "can we add data without retraining" matters operationally.

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1 generate_phase (generic function with 1 method)
2 function generate_phase(n::Int, seed::Int, quality_shift::Float64)
3     rng = MersenneTwister(seed)
4     qi = rand(rng, LogNormal(log(600.0), 0.35), n) # initial rate, bbl/d
5     Di = rand(rng, Uniform(0.35, 0.85), n) # initial decline, 1/yr
6     K = rand(rng, LogNormal(log(40.0) + quality_shift, 0.5), n) # permeability, mD
7
8     # Simplified hyperbolic-decline-flavoured EUR model: higher qi and K help,
9     # step initial decline Di hurts, with a mild q1 x K synergy term.
10     log_eur = 0.525 * log(qi / 600.0) + 1.15 * Di *
11         0.30 * log(K / 40.0) + 0.12 * log(K / 40.0) * log(qi / 600.0)
12     noise = rand(rng, Normal(0.0, 0.20), n)
13     eur = exp.(log_eur .+ noise) # Mbbl
14
15     DataFrame(qi=qi, Di=Di, K=K, EUR=eur, LOGEUR=log.(eur))
16 end
```

```
1 (nA = 40, nB = 25)
2
3 begin
4     Random.seed!(2024)
5     phaseA = generate_phase(40; seed=101, quality_shift=0.0) # core area
6     phaseB = generate_phase(25; seed=202, quality_shift=0.55) # step-out area: lower-
7     K rock
8     (nA=nrow(phaseA), nB=nrow(phaseB))
9 end
```



```
1 begin
2     @df phaseA scatter(:K, :EUR, Label="Phase A (core)", xlabel="permeability K (mD)",
3     ylabel="EUR (Mbbl)", yscale=:log10, xscale=:log10, markersize=4, alpha=0.7,
4     size=(650,450))
5     @df phaseB scatter(:K, :EUR, Label="Phase B (step-out)", markersize=4, alpha=0.7,
6     markershape=:diamond)
7 end
```

2. Shared BNN architecture & Turing model

Same idea as notebook 1: 3 inputs → 5 tanh → 5 tanh → 1 linear output, weights flattened into one vector θ . The model function takes 5 prior_mean / prior_cov so it can be reused both for a vague "fit from scratch" prior and for an informative "posterior-of-A-as-prior" distribution.

```
1 # Shared BNN architecture & Turing model
2
3 ## 2. Shared BNN architecture & Turing model
4
5 Same idea as notebook 1: 3 inputs + 5 tanh + 5 tanh + 1 linear output,
6 weights flattened into one vector 'θ'. The model function takes an
7 explicit 'prior_mean' / 'prior_cov' so it can be reused both for a vague
8 "fit from scratch" prior and for an informative "posterior-of-A-as-prior"
9 distribution.
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